Nila Karthikesan

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Education

University of Maryland, College Park — College Park, MD

B.S. in Computer Science, B.A. in Public Policy, and Minor in Engineering Technology — *Aug. 2020 – May 2024*

- **Director of Engineering, Society of Women Engineers:** Led technical workshops creating 50+ projects with JS, React, Node, REST APIs, and cloud-native backends.

Georgia Institute of Technology — Online

M.S. in Computer Science (Specialization: Artificial Intelligence) — *Aug. 2025 – Present*

- Researching large-scale 3D reconstructions and maintainer of the widely-used open-source repository, contributing to large-scale 3D reconstruction, multi-view geometry, and scalable scene understanding GTSfM
- Co-authoring a CVPR paper ("Building Rome in a Minute") with Prof. Frank Dellaert (former Chief Research Scientist at Skydio), focusing on the scalability of multi-camera reconstruction pipelines on large video-sequence datasets.

Experience

Software Engineer, Platform Infrastructure — Oct. 2025 – Present

Pinterest — *New York, NY*

- Building internal cloud governance and automation for the **Security Platform (PlatSec)** org, managing and securing **thousands of GCP projects and AWS resources** via **Terraform IaC**.
- Designed an **S3-backed Terraform remote state** and workspace strategy with label→tfvars and declarative import automation, enabling safe migration of existing projects with **no-op plans** and no destructive drift.
- Implementing centralized org audit logging (**GCP log sinks → Pub/Sub → Cloud Function → S3 → Gravwell**) for low-latency, scalable security analytics across distributed systems.

Software Engineer — May 2024 – Oct. 2025

Geico — *New York, NY*

- Led development of a **full-stack internal platform** managing GEICO's server inventory, exposing **100,000+ server records** via a React UI and **GraphQL API** to **500+ engineers**.
- Reduced server query by **40%** through improved data modeling, caching, API design across PostgreSQL, GraphQL resolvers.
- Migrated reporting workflows from **Power BI to PostgreSQL/Python**, cutting data processing time by **50%** and improving reliability of infra telemetry pipelines.

Software Engineering Fellow — Dec. 2023 – Jan. 2024

Palantir Technologies — *New York, NY*

- Developed a backend for a proxy service handling **10,000+ IPs** & **100+ monthly users** on an AI platform, focusing on reliable APIs and observability in production.
- Improved project load time by **80%** (minutes → seconds) by containerizing services with **Docker & Kubernetes** and optimizing I/O paths.

Software Engineering Intern — May 2023 – Aug. 2023

Comcast — *Washington, DC*

- Built a **full-stack content moderation platform** using **React.js, Flask, and OpenAI's CLIP**, integrating ML models in production to classify and flag unsafe media across creator content.
- Implemented containerized microservices with **Docker & Kubernetes on AWS**, improving horizontal scalability by **40%** and supporting concurrent moderation workloads.
- Integrated **Elasticsearch** to power low-latency search and review of flagged items, achieving an **8x reduction in retrieval time** and boosting moderator efficiency.

Projects

CLIP-Based Content Moderation System | *Python, FastAPI, Supabase, OpenAI CLIP, Docker, React* — 2024

- Developed a full-stack ML system analyzing video/image embeddings with **OpenAI's CLIP** to detect unsafe content, using **FastAPI, Supabase/Postgres, and Dockerized microservices** for scalable moderation pipelines.
- Built cosine-similarity-based embedding search and React dashboard for reviewers, enabling real-time classification, flagging, and explainable model-inference UX.

Search Engine Optimization Evaluation Agent | *Python, LangChain, OpenAI, Supabase, Streamlit* — 2025

- Created an **LLM-based evaluation agent** using LangChain to generate and rank responses from **ChatGPT/Claude/Perplexity**, storing structured metrics in **Supabase** for analysis and iteration.
- Designed a **Streamlit dashboard** to visualize rankings and quality scores, showcasing agentic orchestration similar to search/recommendation model tuning workflows.

Technical Skills

Languages: Python, Java, Go, C/C++, SQL (Postgres), JavaScript, Rust, R, HTML/CSS
Frameworks: React, Node.js, Flask, FastAPI, GraphQL, Temporal, JUnit
Cloud & Infra: AWS (EC2, S3, RDS, Lambda), GCP (Logging, Pub/Sub, Cloud Functions), Terraform, Docker, Kubernetes, Git
Libraries: Pandas, NumPy, Matplotlib, SciPy